

Alternating Current

Question1

A coil of inductive reactance $\frac{1}{\sqrt{3}} \Omega$ and a resistance 1Ω are connected in series to a 200 V, 50 Hz AC source. The time lag between voltage and current is

AP EAPCET 2025 - 26th May Morning Shift

Options:

A.

$$\frac{1}{1200} \text{ s}$$

B.

$$\frac{1}{600} \text{ s}$$

C.

$$\frac{1}{400} \text{ s}$$

D.

$$\frac{1}{800} \text{ s}$$

Answer: B

Solution:

First, find the angle by which the current lags the voltage. This angle is called ϕ .

Find $\tan \phi$:

The formula is $\tan \phi = \frac{X_L}{R}$, where X_L is the inductive reactance and R is resistance.

Here, $X_L = \frac{1}{\sqrt{3}} \Omega$ and $R = 1 \Omega$.

$$\text{So, } \tan \phi = \frac{\frac{1}{\sqrt{3}}}{1} = \frac{1}{\sqrt{3}}.$$

We know that $\tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$. So, $\phi = \frac{\pi}{6}$.

Find the time period T of the AC supply:

The frequency $f = 50$ Hz, so $T = \frac{1}{f} = \frac{1}{50}$ seconds.

Find the time lag Δt :

The time lag is given by $\Delta t = \frac{\phi}{2\pi} \times T$.

Put the values in: $\Delta t = \frac{\frac{\pi}{6}}{2\pi} \times \frac{1}{50}$.

Simplify it: $\Delta t = \frac{1}{12} \times \frac{1}{50} = \frac{1}{600}$ seconds.

Question2

In a series LCR circuit, the voltages across the capacitor, resistor and inductor are in the ratio 2 : 3 : 6, if the voltage of the AC source in the circuit is 240 V , then the voltage across the inductor is

AP EAPCET 2025 - 26th May Evening Shift

Options:

A.

240 V

B.

144 V

C.

96 V

D.

288 V

Answer: D

Solution:

The voltages across the capacitor (V_C), resistor (V_R), and inductor (V_L) have the ratio 2:3:6.

This means $V_C = 2k$, $V_R = 3k$, and $V_L = 6k$ for some number k .

The total voltage across the circuit is given by: $V = \sqrt{V_R^2 + (V_L - V_C)^2}$ This formula is used because in an LCR circuit, resistor voltage is in phase with supply, while inductor and capacitor voltages are 180° out of phase with each other.

We are told the source voltage is 240 V: $240 = \sqrt{(3k)^2 + (6k - 2k)^2}$

Simplifying inside the square root: $(3k)^2 = 9k^2$, $(6k - 2k)^2 = (4k)^2 = 16k^2$ So,
 $240 = \sqrt{9k^2 + 16k^2} = \sqrt{25k^2} = 5k$

Now solve for k : $5k = 240 \implies k = \frac{240}{5} = 48$

Now, calculate the voltage across the inductor: $V_L = 6k = 6 \times 48 = 288 \text{ V}$

Question3

If the voltage and current in an AC circuit are respectively $50 \sin(50t) \text{ V}$ and $50 \sin(50t + \frac{\pi}{4}) \text{ mA}$, then the power dissipated in the circuit is nearly

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Options:

A.

1.296 W

B.

0.648 W

C.

0.884 W

D.

1.768 W

Answer: C

Solution:

The formula to find the average (real) power in an AC circuit is:

$$P = V_{\text{rms}} \times I_{\text{rms}} \times \cos \phi \text{ where } \phi \text{ is the phase difference between voltage and current.}$$

The voltage is $50 \sin(50t)$, so its maximum (peak) value is 50 V.

The current is $50 \sin(50t + \frac{\pi}{4})$, so its peak value is $50 \text{ mA} = 50 \times 10^{-3} \text{ A}$.

The phase difference ϕ between voltage and current is $\frac{\pi}{4}$.

To get the RMS (root mean square) values from the peak values, divide by $\sqrt{2}$:

$$V_{\text{rms}} = \frac{50}{\sqrt{2}}$$

$$I_{\text{rms}} = \frac{50 \times 10^{-3}}{\sqrt{2}}$$

Now, put these values into the power formula:

$$P = \frac{50}{\sqrt{2}} \times \frac{50 \times 10^{-3}}{\sqrt{2}} \times \cos\left(\frac{\pi}{4}\right)$$

$$\cos\left(\frac{\pi}{4}\right) \text{ is } \frac{1}{\sqrt{2}}.$$

So,

$$P = \frac{50}{\sqrt{2}} \times \frac{50}{\sqrt{2}} \times 10^{-3} \times \frac{1}{\sqrt{2}}$$

Multiply the numbers:

$$= \frac{50 \times 50 \times 10^{-3}}{\sqrt{2} \times \sqrt{2} \times \sqrt{2}}$$

$$\sqrt{2} \times \sqrt{2} \times \sqrt{2} = 2\sqrt{2}$$

$$= \frac{2500 \times 10^{-3}}{2\sqrt{2}}$$

$$= \frac{2.5}{2\sqrt{2}}$$

This is approximately equal to 0.884 W.

Question4

In a series resonant LCR circuit, for the power dissipated to become half of the maximum power dissipated, the current amplitude is

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Options:

A.

$\frac{1}{\sqrt{2}}$ times its maximum value.

B.

$1/2$ times its maximum value.

C.

twice its maximum value.

D.

$\sqrt{2}$ times its maximum value.

Answer: A

Solution:

Given:

In a **series LCR circuit at resonance**, the **power dissipated** is maximum.

We need to find:

For the **power dissipated** to become **half of its maximum value**, what is the **current amplitude**?

Step 1: Relation between Power and Current

The average power dissipated in an AC circuit is given by:

$$P = I_{\text{rms}}^2 R$$

At resonance, the impedance is purely resistive ($Z = R$), and the current is maximum ($I = I_{\text{max}}$).

So maximum power is:

$$P_{\text{max}} = I_{\text{rms,max}}^2 R = \left(\frac{I_{\text{max}}}{\sqrt{2}} \right)^2 R = \frac{I_{\text{max}}^2 R}{2}$$

But we only need **relative power**, so we can use $P \propto I^2$.

Step 2: Condition for half power

We are told $P = \frac{1}{2}P_{\max}$

So:

$$I^2 R = \frac{1}{2} I_{\max}^2 R$$

Simplify:

$$I = \frac{I_{\max}}{\sqrt{2}}$$

 **Final Answer:**

Option A: $\frac{1}{\sqrt{2}}$ times its maximum value.

So the current amplitude becomes $1/\sqrt{2}$ times its maximum value when the power is half of the maximum.

Question5

If a resistor of resistance 4Ω , a capacitor of capacitive reactance 6Ω and an inductor of inductive reactance 9Ω are connected in series with an AC source, then the impedance of the circuit is

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Options:

A.

19Ω

B.

11Ω

C.

7Ω

D.

5Ω

Answer: D

Solution:

Impedance,

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$
$$= \sqrt{4^2 + (9 - 6)^2} = \sqrt{16 + 9} = 5\Omega$$

Question6

A resistor of 450Ω and an inductor are connected in series to an AC source of frequency $\frac{75}{\pi}$ Hz. If the power factor of the circuit is 0.6 , then the inductance connected in the circuit is

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Options:

A.

6 mH

B.

4 H

C.

4 mH

D.

6 H

Answer: B

Solution:

Power factor

$$\cos \phi = \frac{R}{Z}$$

$$0.6 = \frac{R}{\sqrt{R^2 + X_L^2}}$$

$$\Rightarrow 0.36 = \frac{R^2}{R^2 + (2\pi fL)^2}$$

$$\Rightarrow 0.36 = \frac{(450)^2}{(450)^2 + \left(2\pi \times \frac{75}{\pi} L\right)^2}$$

$$\Rightarrow 0.36 = \frac{(450)^2}{450^2 + (150L)^2}$$

Solving, we get

$$L = 4H$$

Question7

For better tuning of a series LCR circuit in a communication system, the preferred combination is

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Options:

A.

$$R = 20\Omega : L = 15H : C = 35\mu F$$

B.

$$R = 15\Omega : L = 40H : C = 20\mu F$$

C.

$$R = 25\Omega : L = 15H : C = 45\mu F$$

D.

$$R = 15\Omega : L = 20H : C = 45\mu F$$

Answer: B

Solution:

For better tuning of a series LCR circuit in a communication system. Quality factor ' Q ' should be high.

i.e.

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

So, for high value of Q , value of R and C should be minimum and value L should be maximum.

Question8

In an LCR series circuit, if the potential differences across inductor, capacitor and resistor are 60 V, 30 V and 40 V respectively, then the AC voltage applied to the circuit is

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Options:

A.

50 V

B.

70 V

C.

130 V

D.

60 V

Answer: A

Solution:

AC voltage applied to the circuit,

$$\begin{aligned} V &= \sqrt{V_R^2 + (V_L - V_C)^2} \\ &= \sqrt{40^2 + (60 - 30)^2} \\ &= \sqrt{1600 + 900} = \sqrt{2500} \\ &= 50 \text{ V} \end{aligned}$$

Question9

The resonant frequency of an LC circuit is f_0 . If a dielectric slab of constant 16 is inserted completely between the plates of the capacitor, then the resonant frequency is

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Options:

A.

$$\frac{f}{2}$$

B.

$$26$$

C.

$$\frac{6}{4}$$

D.

$$4f$$

Answer: C

Solution:

Resonance frequency

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

When dielectric material is introduced, then new capacitance,

$$C' = C_0 K = 16C_0 = 16C$$

$\therefore$ New resonance frequency

$$\begin{aligned} f'_0 &= \frac{1}{2\pi\sqrt{LC'}} = \frac{1}{2\pi\sqrt{L \times 16C}} \\ &= \frac{1}{4} \times \frac{1}{2\pi\sqrt{LC}} = \frac{f_0}{4} \end{aligned}$$

Question10

An alternating current is given by $i = (3 \sin \omega t + 4 \cos \omega t)$ A. The rms current will be

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Options:

A. $\frac{7}{\sqrt{2}}$ A

B. $\frac{1}{\sqrt{2}}$ A

C. $\frac{5}{\sqrt{2}}$ A

D. $\frac{3}{\sqrt{2}}$ A

Answer: C

Solution:

The rms value is found from

writing $i(t) = A \sin \omega t + B \cos \omega t$ with $A = 3$, $B = 4$,

using

$$i_{\text{rms}} = \sqrt{\langle i^2 \rangle} = \sqrt{\frac{A^2 + B^2}{2}} = \frac{\sqrt{3^2 + 4^2}}{\sqrt{2}} = \frac{5}{\sqrt{2}} \text{ A}.$$

So the correct choice is Option C: $\frac{5}{\sqrt{2}}$ A.

Question11

A 50 Hz AC circuit has a 10 mH inductor and a 2Ω resistor in series. The value of capacitance to be placed in series in the circuit to make the circuit power factor as unity is

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Options:

A. $1.014 \times 10^{-6} \text{ F}$

B. $1.014 \times 10^{-3} \text{ F}$

C. $2.6 \times 10^{-3} \text{ F}$

D. $4.125 \times 10^{-3} \text{ F}$

Answer: B

Solution:

Given:

Frequency, $f = 50 \text{ Hz}$

Inductance, $L = 10 \text{ mH} = 10 \times 10^{-3} \text{ H}$

Resistance, $R = 2 \Omega$

First, calculate the inductive reactance (X_L) using the formula:

$$X_L = 2\pi fL$$

Substituting the given values:

$$X_L = 2\pi \times 50 \times 10 \times 10^{-3} = \pi$$

For the circuit to have a unit power factor, the inductive reactance (X_L) must equal the capacitive reactance (X_C). Therefore,

$$X_C = X_L$$

The formula for capacitive reactance is:

$$X_C = \frac{1}{2\pi fC}$$

Setting $X_C = X_L$, we have:

$$\pi = \frac{1}{2\pi \times 50 \times C}$$

Solving for C :

$$C = \frac{1}{2(\pi^2) \times 50}$$

$$C = \frac{1}{2 \times 9.8696 \times 50}$$

$$C = \frac{1}{973.036}$$

$$C = 1.014 \times 10^{-3} \text{ F} = 1.014 \text{ mF}$$

Thus, the required capacitance is $1.014 \times 10^{-3} \text{ F}$.

Question12

A resistor of resistance R , inductor of inductive reactance $2R$ and a capacitor of capacitive reactance X_C are connected in series to an AC source. If the series $L - C - R$ circuit is in resonance, then the power factor of the circuit and the value, X_C are respectively

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Options:

A. 0.5 and $4R$

B. 1 and $2R$

C. 0.5 and $2R$

D. 1 and $4R$

Answer: B

Solution:

Given an $L - C - R$ circuit in resonance, let's analyze it:

Provided Values:

Inductive Reactance (X_L): $2R$

Resistance (R): R

Resonance Condition:

In a resonant circuit, the inductive reactance is equal to the capacitive reactance. Therefore:

$$X_L = X_C$$

$$X_C = 2R$$

Power Factor:

The power factor of a circuit is calculated as:

$$P = \frac{\text{Resistance}}{\text{Impedance (Z)}}$$

The impedance in an $L - C - R$ circuit, when in resonance and when $X_L = X_C$, is:

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Since $X_L = X_C$ in resonance:

$$Z = \sqrt{R^2 + (2R - 2R)^2} = R$$

Thus, the power factor is:

$$P = \frac{R}{R} = 1$$

Therefore, when the circuit is in resonance, the capacitive reactance $X_C = 2R$, and the power factor is 1.

Question13

An inductor and a resistor are connected in series to an AC source of voltage $144 \sin \left(100\pi t + \frac{\pi}{2}\right)$ V. If the current in the circuit is $6 \sin \left(100\pi t + \frac{\pi}{6}\right)$ A, then the resistance of the resistor is

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Options:

A. 24Ω

B. 36Ω

C. 12Ω

D. 18Ω

Answer: C

Solution:

Given:

The voltage expression: $V = 144 \sin \left(100\pi t + \frac{\pi}{2}\right)$ V

The current expression: $I = 6 \sin \left(100\pi t + \frac{\pi}{6}\right)$ A

Analysis

Standard Form Comparison:

The standard forms of voltage and current are:

$$V = V_0 \sin(\omega t + \phi_v)$$

$$I = I_0 \sin(\omega t + \phi_i)$$

Phase Difference:

Comparing the given equations, we identify the phases:

$$\text{Voltage phase: } \phi_v = \frac{\pi}{2}$$

$$\text{Current phase: } \phi_i = \frac{\pi}{6}$$

The phase difference, ϕ , between voltage and current is:

$$\phi = \phi_v - \phi_i = \frac{\pi}{2} - \frac{\pi}{6} = \frac{\pi}{3} = 60^\circ$$

Impedance Calculation:

The impedance Z of the circuit is given by:

$$Z = \frac{V_0}{I_0} = \frac{144}{6} = 24 \Omega$$

Resistance Calculation:

The relationship between impedance Z , resistance R , and the phase angle ϕ is given by:

$$\frac{R}{Z} = \cos \phi$$

Solving for R , we have:

$$R = \cos(60^\circ) \times Z = \frac{1}{2} \times 24 = 12 \Omega$$

Therefore, the resistance of the resistor is 12Ω .

Question14

A resistance of 20Ω is connected to a source of an alternating potential $V = 200 \sin(10\pi t)$. If t is the time taken by the current to change from the peak value to rms value, then t is (in seconds)

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Options:

A. 25×10^{-1}

B. 2.5×10^{-4}

C. 25×10^{-2}

D. 2.5×10^{-2}

Answer: D

Solution:

Given:

Resistance: $R = 20 \Omega$

Alternating potential: $V = 200 \sin(10\pi t)$

The expression for the alternating current I can be derived as follows:

$$I = \frac{V}{R} = \frac{200 \sin(10\pi t)}{20} = 10 \sin(10\pi t)$$

We know that the root mean square (RMS) current is related to the peak current by the equation:

$$I_{\text{RMS}} = \frac{I_{\text{peak}}}{\sqrt{2}}$$

Thus, the RMS current is also expressed as:

$$I_{\text{RMS}} = I_{\text{peak}} \sin(10\pi t)$$

Setting the RMS and the peak relation,

$$\frac{I_{\text{peak}}}{\sqrt{2}} = I_{\text{peak}} \sin(10\pi t)$$

This simplifies to:

$$\frac{1}{\sqrt{2}} = \sin(10\pi t)$$

The sine of $\frac{\pi}{4}$ is known to be $\frac{1}{\sqrt{2}}$, so:

$$\sin\left(\frac{\pi}{4}\right) = \sin(10\pi t) \Rightarrow 10\pi t = \frac{\pi}{4}$$

Solving for t :

$$t = \frac{1}{40} \Rightarrow t = 2.5 \times 10^{-2} \text{ s}$$

Question15

A full wave rectifier circuit is operating from 50 Hz mains. The fundamental frequency in the ripple output will be

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Options:

A. 50 Hz

B. 70.7 Hz

C. 100 Hz

D. 25 Hz

Answer: C

Solution:

In a full-wave rectifier, both the positive and negative halves of the AC input are used, effectively doubling the frequency of the output ripple compared to the mains frequency. Here's a step-by-step explanation:

The mains frequency is 50 Hz.

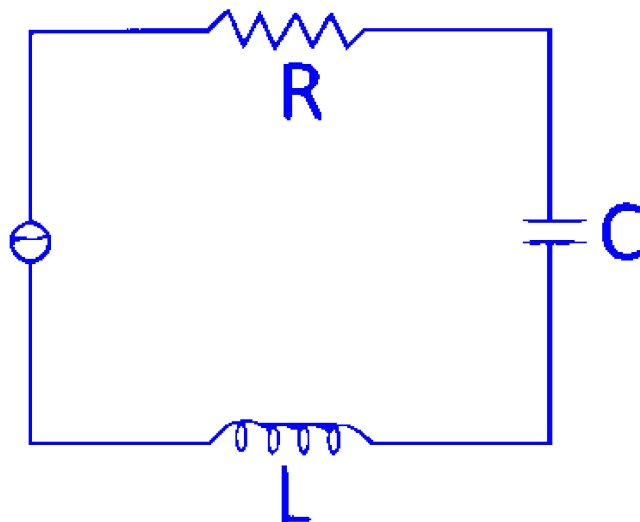
The full-wave rectifier inverts the negative half cycle, so the ripple frequency becomes:

$$f_{\text{ripple}} = 2 \times 50 \text{ Hz} = 100 \text{ Hz}.$$

Therefore, the fundamental frequency in the ripple output is 100 Hz, which is Option C.

Question16

A series $L - C - R$ circuit is shown in the figure. Where the inductance of 10 H , capacitance $40\mu\text{ F}$ and resistance 60Ω are connected to a variable frequency 240 V source. The current at resonating frequency is



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Options:

A. 4 A

B. $2A$

C. $5.4 A$

D. $5.8 A$

Answer: A

Solution:

In the given LCR circuit, the components have the following values:

Inductance, $L = 10 \text{ H}$

Capacitance, $C = 40 \mu\text{F} = 4 \times 10^{-5} \text{ F}$

Resistance, $R = 60 \Omega$

The circuit is connected to a variable frequency source of 240 V .

To find the resonant frequency (ω_r), we use the formula:

$$\omega_r = \frac{1}{\sqrt{LC}}$$

Substituting the given values:

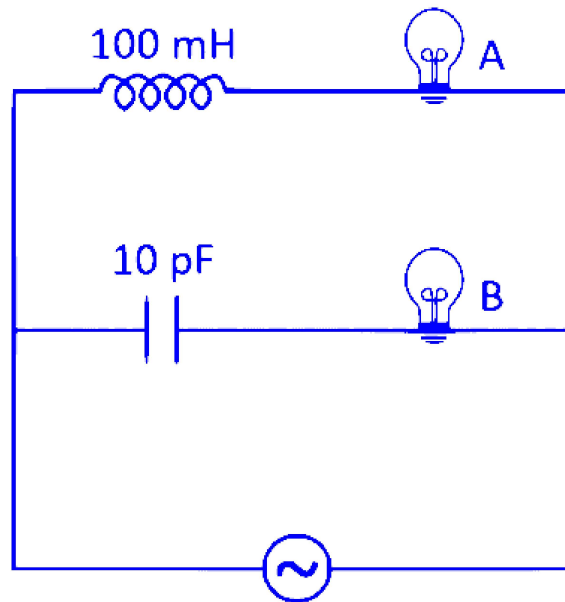
$$\omega_r = \frac{1}{\sqrt{10 \times 4 \times 10^{-5}}} = \frac{1}{2 \times 10^{-2}} = 50 \text{ rad/s}$$

At resonance, the impedance Z of the circuit is minimized to the resistance R only. Thus, the current I at resonant frequency can be calculated by:

$$I = \frac{V}{Z} = \frac{V}{R} = \frac{240}{60} = 4 \text{ A}$$

Question17

In the figure. If A and B are identical bulbs, which bulb glows brighter.



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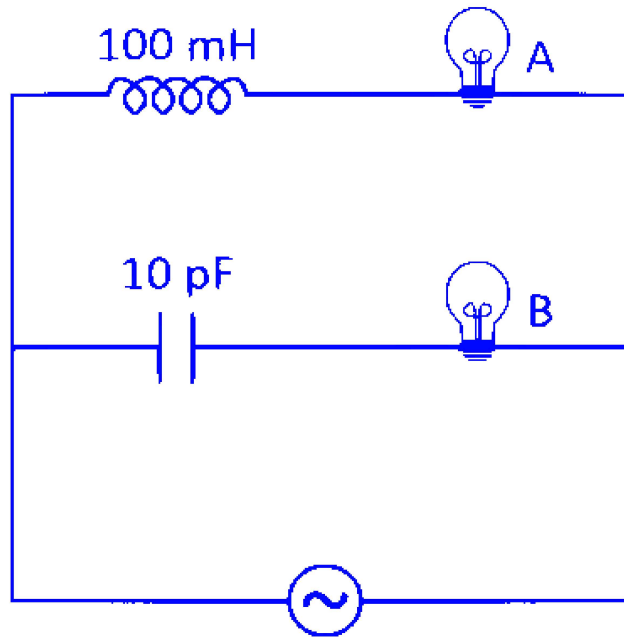
Options:

- A. A
- B. B
- C. Both with equal brightness
- D. Both do not glow

Answer: A

Solution:

We know that, $X_C = 1/\omega C$ and $X_L = \omega L$



According to figure,

$$C = 10\text{pF} = 10 \times 10^{-12} \text{ F}$$

$$L = 100\text{mH} = 0.1\text{H}$$

Here, $C \ll L$

$$\text{So, } X_C \gg X_L$$

$$\therefore i_A \gg i_B$$

Hence, bulb A will glow brighter.

Question18

An $R - L - C$ circuit consists of a 150Ω resistor, $20\mu\text{F}$ capacitor and a 500 mH inductor connected in series with a 100 V AC supply. The angular frequency of the supply voltage is 400rads^{-1} . The phase angle between current and the applied voltage is

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Options:

A. $\tan^{-1}(0.8)$

B. $\tan^{-1}(0.25)$

C. $\tan^{-1}(0.6)$

D. $\tan^{-1}(0.5)$

Answer: D

Solution:

To determine the phase angle ϕ between the current and the applied voltage in an $R - L - C$ circuit, we need to use the following relationship:

$$\tan \phi = \frac{X_L - X_C}{R}$$

where:

- X_L is the inductive reactance
- X_C is the capacitive reactance
- R is the resistance

Given:

- Resistance, $R = 150\Omega$
- Capacitance, $C = 20\mu\text{F} = 20 \times 10^{-6}\text{F}$
- Inductance, $L = 500\text{mH} = 500 \times 10^{-3}\text{H}$
- Angular frequency, $\omega = 400\text{rad s}^{-1}$

The inductive reactance X_L is calculated as:

$$X_L = \omega L$$

Substituting the values:

$$X_L = 400 \times 500 \times 10^{-3}$$

$$X_L = 200\Omega$$

The capacitive reactance X_C is calculated as:

$$X_C = \frac{1}{\omega C}$$

Substituting the values:

$$X_C = \frac{1}{400 \times 20 \times 10^{-6}}$$

$$X_C = 125\Omega$$

Now, we can find the phase angle using the equation:

$$\tan \phi = \frac{X_L - X_C}{R}$$

Substituting the values:

$$\tan \phi = \frac{200-125}{150}$$

$$\tan \phi = \frac{75}{150}$$

$$\tan \phi = 0.5$$

Therefore, the phase angle ϕ is:

$$\phi = \tan^{-1}(0.5)$$

So, the correct answer is:

$$\text{Option D } \tan^{-1}(0.5)$$

Question19

Capacitive reactance of a capacitor in an AC circuit is $3\text{k}\Omega$. If this capacitor is connected to a new AC source of double frequency, the capacitive reactance will become.

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Options:

A. $1.5\text{k}\Omega$

B. $3\text{k}\Omega$

C. $6\text{k}\Omega$

D. $5.2\text{k}\Omega$

Answer: A

Solution:

Given capacitive reactance,

$$X_{C_1} = 3\text{k}\Omega$$

we know that,

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

$$\begin{aligned}\Rightarrow X_C &\propto \frac{1}{f} \Rightarrow \frac{X_{C_2}}{X_{C_1}} = \frac{f_1}{f_2} \\ &= \frac{f_1}{2f_1} = \frac{1}{2} \quad (\because f_2 = 2f_1) \\ \Rightarrow X_{C_2} &= \frac{X_{C_1}}{2} \\ &= \frac{3\text{k}\Omega}{2} = 1.5\text{k}\Omega\end{aligned}$$

Question20

A coil of inductance 0.1 H and resistance 110Ω is connected to a source of 110 V and 350 Hz . The phase difference between the voltage maximum and the current maximum is

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Options:

- A. $\tan^{-1}(1.5)$
- B. $\tan^{-1}(0.5)$
- C. $\tan^{-1}(1.73)$
- D. $\tan^{-1}(2)$

Answer: D

Solution:

Given, Inductance, $L = 0.1\text{ H}$

Resistance, $R = 110\Omega$;

Potential difference, $V = 110\text{ V}$

Frequency, $f = 350\text{ Hz}$

Phase difference between maximum voltage and maximum current is given as

$$\begin{aligned}\tan \phi &= \frac{X_L}{R} \\ &= \frac{2\pi fL}{R} = \frac{2 \times \frac{22}{7} \times 350 \times 01}{110} = 2 \\ \Rightarrow \tan \phi &= 2 \\ \Rightarrow \phi &= \tan^{-1}(2)\end{aligned}$$

Question21

A 20 V AC is applied to a circuit consisting of a resistor and a coil with negligible resistance. If the voltage across the resistor is 12 V, the voltage across the coil is

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Options:

A. 16 V

B. 10 V

C. 8 V

D. 6 V

Answer: A

Solution:

Given, $V_{\text{rms}} = 20 \text{ V}$

Voltage across resistance = 12 V

Let voltage across inductor be V_L

As we know that,

$$\begin{aligned}V^2 &= V_R^2 + V_L^2 \\ \Rightarrow V_L^2 &= V^2 - V_R^2 = (20)^2 - (12)^2 \\ &= 400 - 144 = 256 \text{ V} \\ V_L &= 16 \text{ V}\end{aligned}$$

Question22

A bulb of resistance 280Ω is supplied with a 200 V AC supply. What is the peak current?

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Options:

- A. Nearly 1 A
- B. Nearly 2 A
- C. Nearly 1.4 A
- D. Nearly 2.8 A

Answer: A

Solution:

Given, resistance of bulb, $R = 280\Omega$

Supply voltage $V_{\text{rms}} = 200 \text{ V}$

As, $V_{\text{rms}} = \frac{V_0}{\sqrt{2}} = \text{root mean square speed}$

$$\Rightarrow V_0 = \sqrt{2}V_{\text{rms}}$$

where, V_0 is peak voltage,

As, $V = IR$

$$\therefore I_0 = \frac{V_0}{R} = \frac{\sqrt{2} \times 200}{280} = \frac{5\sqrt{2}}{7} = 1.010 \text{ A}$$

Which is nearly 1A.

Question23

A resonant frequency of a current is f . If the capacitance is made four times the initial value, then the resonant frequency will become

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Options:

A. $\frac{f}{2}$

B. $2f$

C. f

D. $\frac{f}{4}$

Answer: A

Solution:

Given,

Initial resonance frequency, $f_i = f$

Initial capacitance and inductance be $C_1 = C$ and $L_1 = L$

Final capacitance and inductance one $C_2 = 4C$ and $L_2 = L$

Let final frequency be f' .

Since, $\omega = \frac{1}{\sqrt{LC}} = 2\pi f$

where, ω and f and angular frequency and frequency.

$$\therefore f \propto \frac{1}{\sqrt{LC}} \Rightarrow \frac{f}{f'} = \sqrt{\frac{L'C'}{LC}} = \frac{\sqrt{L4C}}{\sqrt{LC}} = \sqrt{4} = 2$$

$$\therefore f' = \frac{f}{2}$$
